MEDIPULSE

Healthcare Admissions & Analytics Dashboard

PROJECT REPORT

Project Domain	Healthcare Data Analytics
Tools Used	Excel • Power Query • Power BI • DAX
Reporting Period	January 2019 – December 2024

1. Project Overview and Objective

MediPulse is a healthcare admissions and analytics project developed using Microsoft Excel, Power Query, and Power BI. It analyses patient demographics, admission patterns, hospital-level activity, and billing information through interactive dashboards.

Objectives: clean and prepare the healthcare data; organize patient/admission and hospital tables; create interactive KPI cards and visualizations; and present descriptive insights that support healthcare reporting.

2. Data Sources

The healthcare dataset used in this project was downloaded from the GitHub repository below:

Dataset source: Python Analysis on Healthcare Dataset – GitHub repository

Project domain: Healthcare Data Analytics – patient admissions, patient demographics, hospital operations, and billing analysis.
Reporting period / years shown in the dashboards: January 2019 – December 2024.
Raw dataset: 2,000 rows and 15 columns. Cleaned dataset: 1,411 rows and 15 columns. A total of 589 records containing one or more blank values were removed during data cleaning.

3. Problem Statement

Raw healthcare admission records can contain missing values and inconsistent text, making it difficult to analyse performance directly. This project prepares the records and presents them in an interactive reporting format.

Key questions: How are admissions distributed by age group and gender? Which medical conditions and admission types appear most often? How do records vary by hospital? How does billing vary by insurance provider, admission type, and medical condition? How can slicers help users explore the data?

4. Attribute (Column / Feature) Details

The original healthcare dataset contains the following fields:

Attribute Name	Data Type	Description
Name	Text	Patient name recorded in the admission dataset.
Age	Whole number	Patient age in years.
Gender	Text / Category	Patient gender category.
Blood Type	Text / Category	Recorded blood group.
Medical Condition	Text / Category	Medical condition associated with the record.
Date of Admission	Date	Date the patient was admitted.
Doctor	Text	Doctor name recorded in the dataset.

Hospital	Text	Hospital name associated with the admission.
Insurance Provider	Text / Category	Insurance provider recorded for the patient.
Billing Amount	Decimal / Currency	Billing amount associated with the record.
Room Number	Whole number	Room number recorded in the dataset.
Admission Type	Text / Category	Admission classification.
Discharge Date	Date	Date the patient was discharged.
Medication	Text / Category	Medication recorded in the dataset.
Test Results	Text / Category	Test result category recorded in the dataset.

Additional model fields: Admission_ID is a generated row-level admission identifier in Patient_Dataset_New. Hospital_ID is a generated key used to connect hospital names across the model.

5. Tools & Technologies

Microsoft Excel: reviewing source data and organizing worksheets. Power Query: cleaning, standardization, filtering, and transformation. Power BI Desktop: data modelling, DAX calculations, visualizations, and interactive dashboards. DAX: measures for KPI cards and report metrics.

6. Data Pre-Processing (Excel / Power Query)

Reviewed the original 15-column dataset. Removed records containing one or more blank values, reducing the dataset from 2,000 to 1,411 records. Standardized text values and naming inconsistencies where identified. Prepared the cleaned healthcare dataset for Power BI.

Created Patient_Dataset_New with one row per admission record and fields including Admission_ID, Patient_Name, Age, and Hospital_ID. Created Hospital_Dataset_New with Hospital_ID and 1,385 unique Hospital_Name values. The original raw dataset was kept separate from the cleaned workbook.

Note: Only confirmed preparation activities are listed. Add duplicate removal or Pivot Table analysis only if these steps were actually completed and verified.

7. Data Modelling and DAX (Power BI)

Main tables: Clean_Dataset_Healthcare_Patient (cleaned admission and billing details); Patient_Dataset_New (admission-level patient details); Hospital_Dataset_New (hospital lookup with Hospital_ID and Hospital_Name).

The model uses Hospital_ID relationships from Hospital_Dataset_New (one side) to Patient_Dataset_New and Clean_Dataset_Healthcare_Patient (many side). Confirm that both relationships are active and that key values match in the final PBIX file.

Example DAX measures (verify exact table/column names in the PBIX before presenting these as the stored formulas):

Total Admissions = COUNTROWS('Clean_Dataset_Healthcare_Patient')

Total Hospitals = DISTINCTCOUNT('Hospital_Dataset_New'[Hospital_ID])

Average Patient Age = AVERAGE('Clean_Dataset_Healthcare_Patient'[Age])

Total Billing Amount = SUM('Clean_Dataset_Healthcare_Patient'[Billing Amount])

Average Billing Amount = AVERAGE('Clean_Dataset_Healthcare_Patient'[Billing Amount])

The admissions gauge displays 2,000 as its target/reference value. Admission_ID is a generated admission-row identifier, not a verified unique patient identifier.

Data Modelling and DAX – Screenshot Evidence

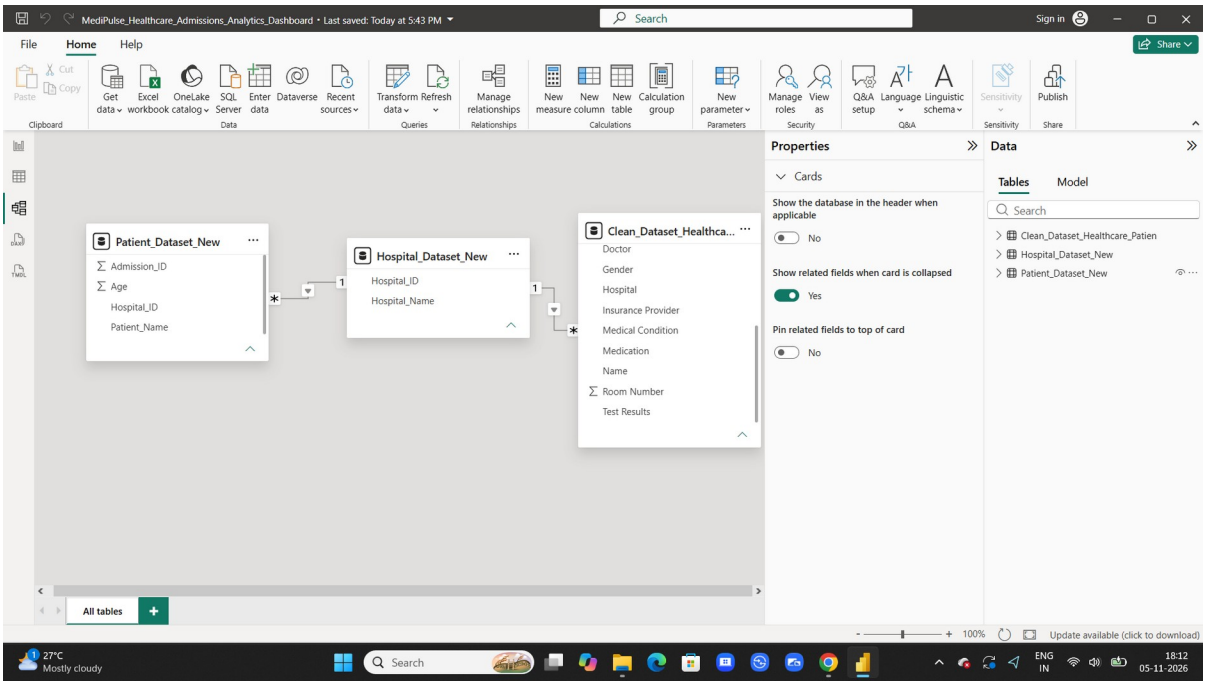


Figure 1: Power BI data model showing the three tables and Hospital_ID relationships.

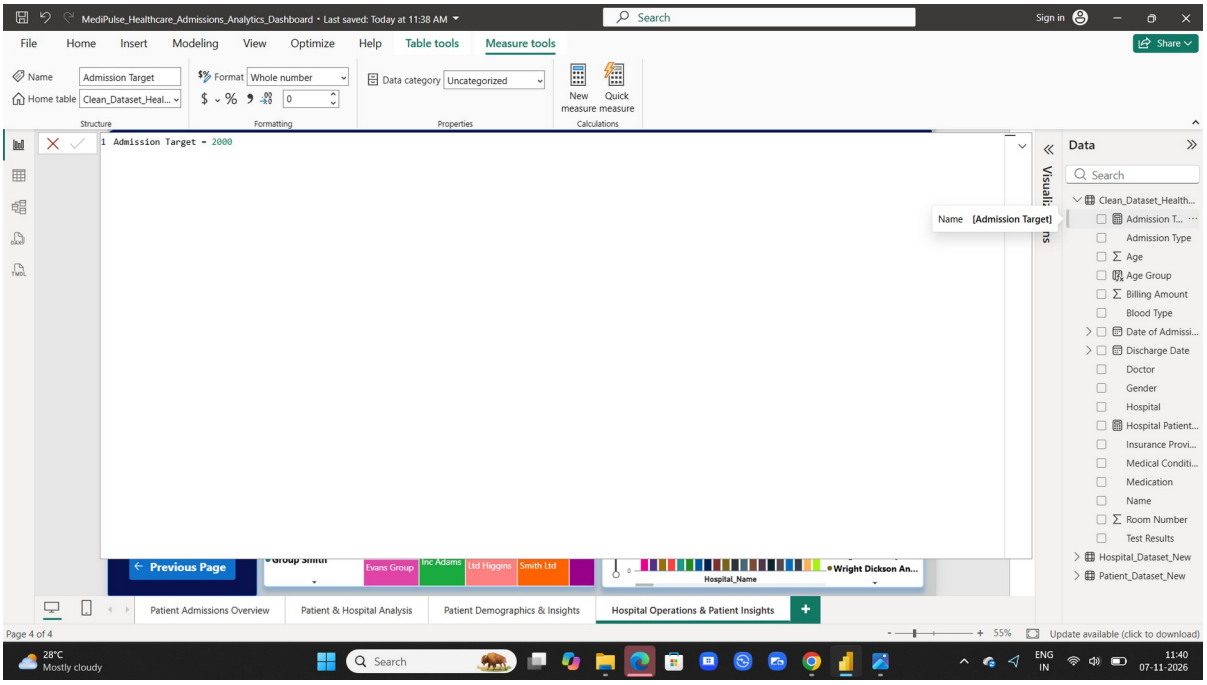


Figure 2: DAX measure – Admission Target (2,000).

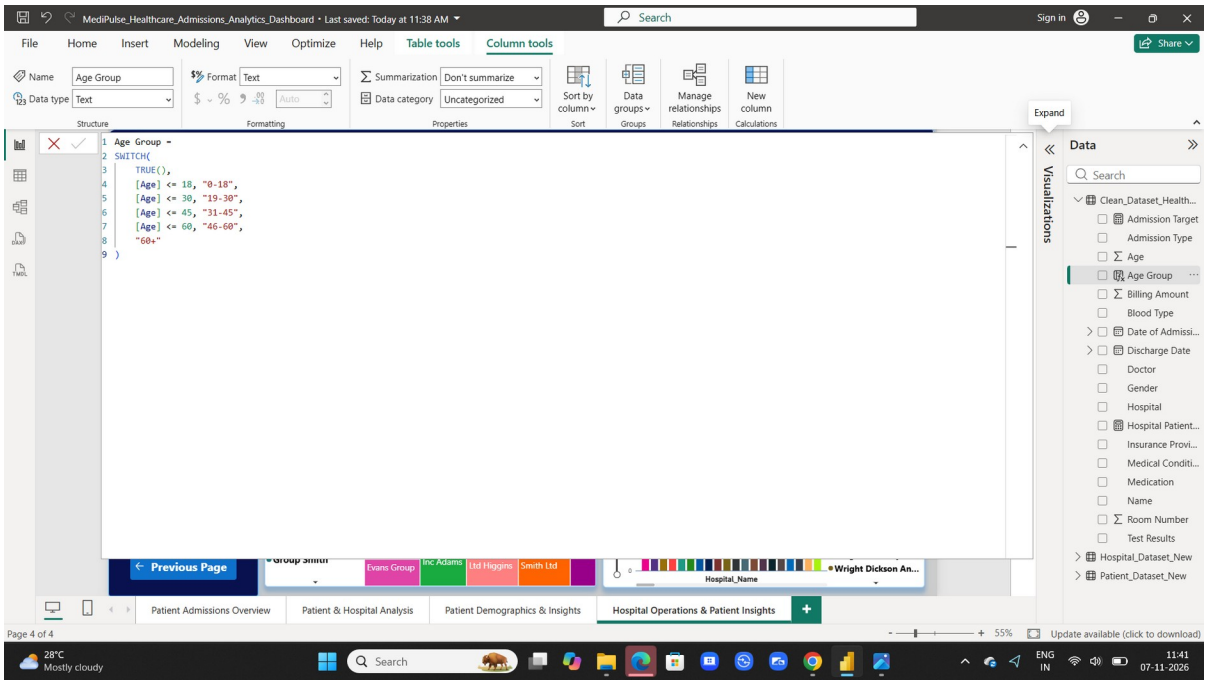


Figure 3: DAX calculated column – Age Group using SWITCH and TRUE().

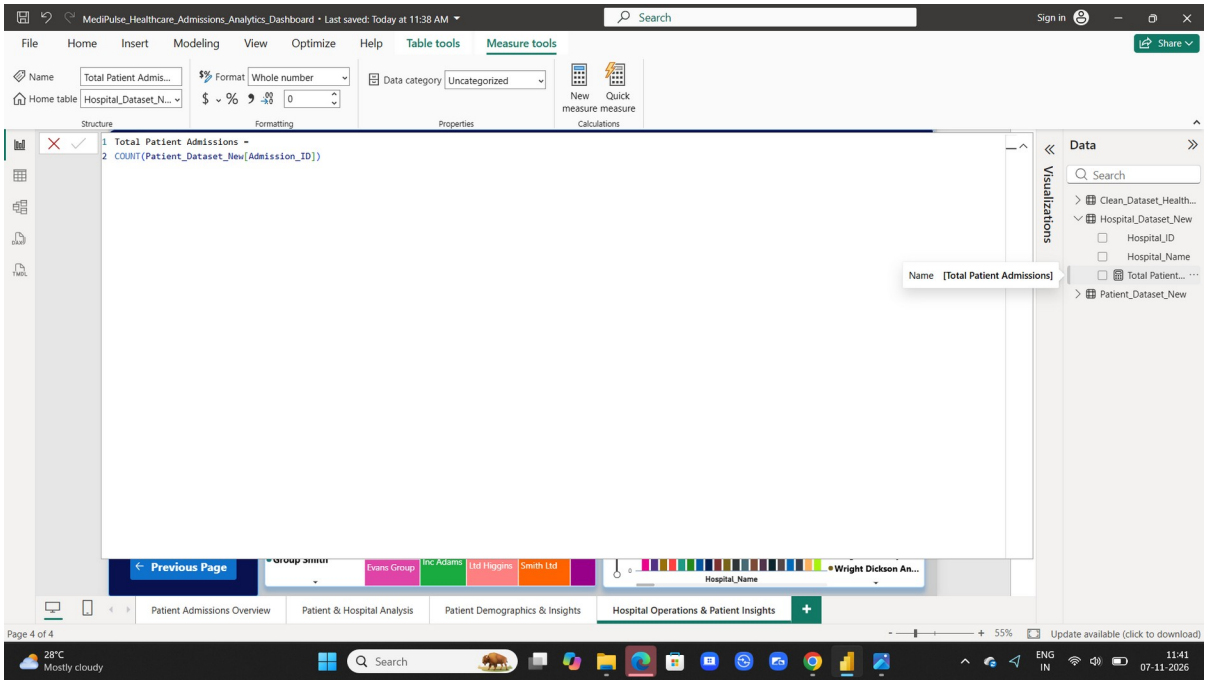


Figure 4: DAX measure – Total Patient Admissions using COUNT().

8. Analysis and Visualizations (Power BI)

Dashboard 1 – Patient Admissions Overview

KPI cards for Total Admissions, Average Patient Age, Male Patients, and Female Patients; Admissions by Gender; Admissions by Age Group; Admission Type; Admissions by Medical Condition. Slicers: Gender, Hospital, Admission Type, Date of Admission.

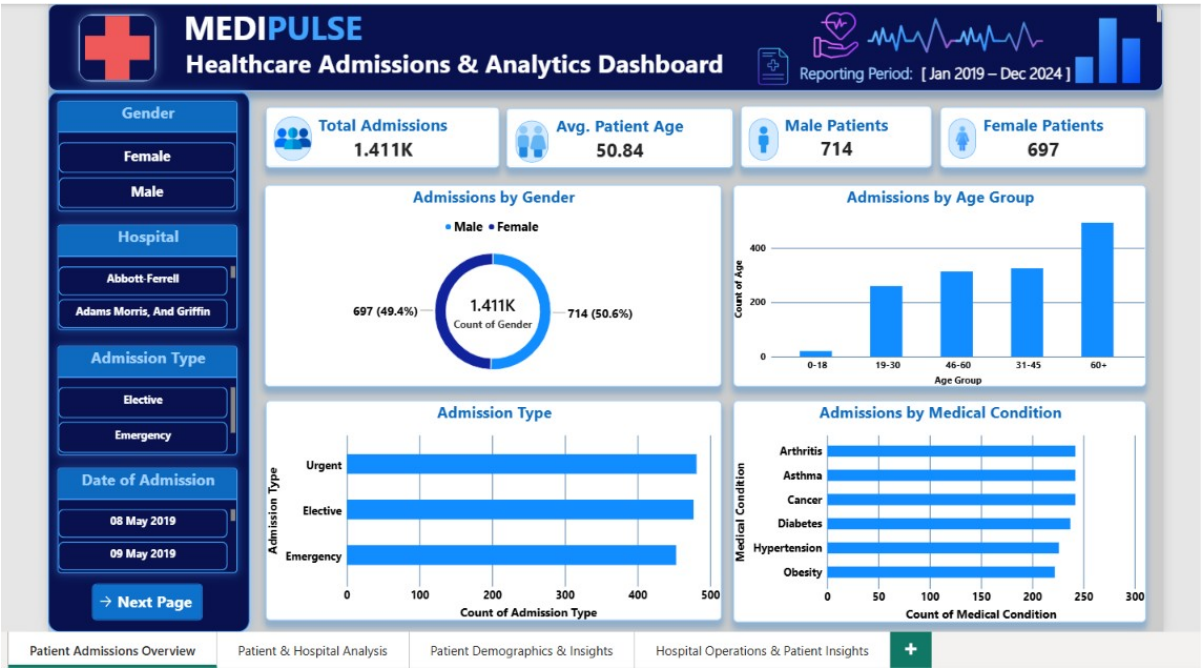


Figure 5: Dashboard 1 – Patient Admissions Overview.

Dashboard 2 – Patient & Hospital Analysis

Total Billing Amount and Average Billing Amount cards; Billing by Insurance Provider; Billing by Admission Type; Billing by Medical Condition; Top 10 Hospitals by Billing Amount. Slicers: Insurance Provider, Medical Condition, Admission Type.

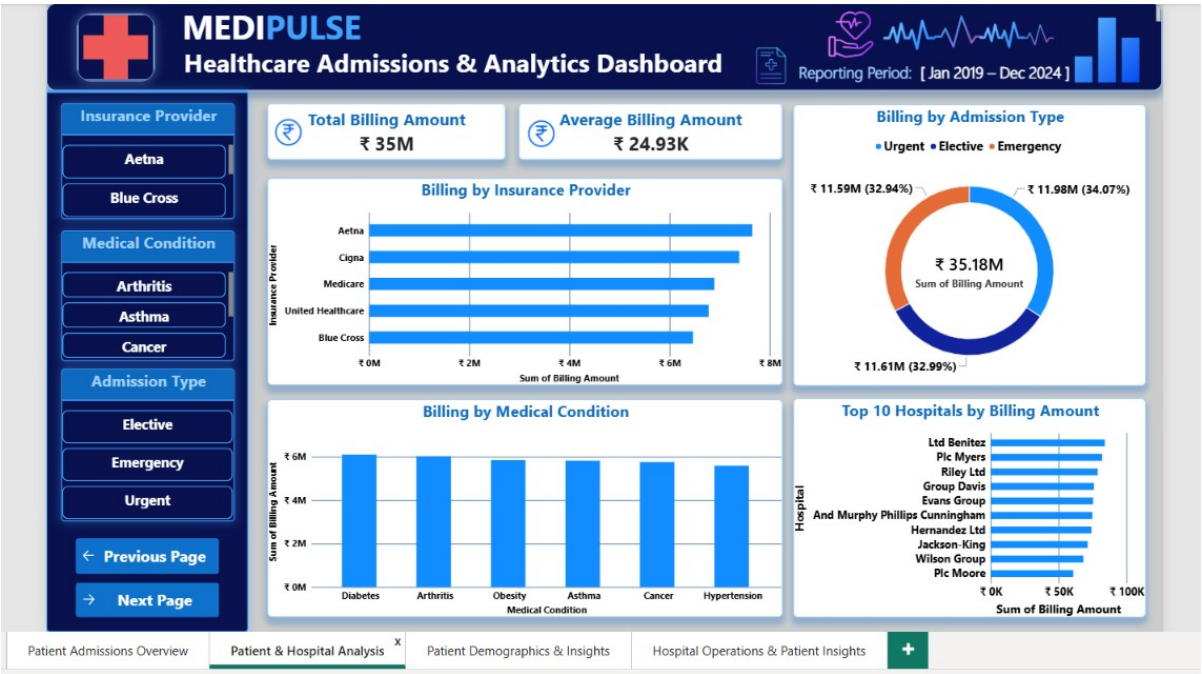


Figure 6: Dashboard 2 – Patient & Hospital Analysis.

Dashboard 3 – Patient Demographics & Insights

Patient Distribution by Age Group; Patient Distribution by Gender; Patient Distribution by Medical Condition; Patient Age vs Billing Analysis. Slicers: Age Group, Gender, Medical Condition.

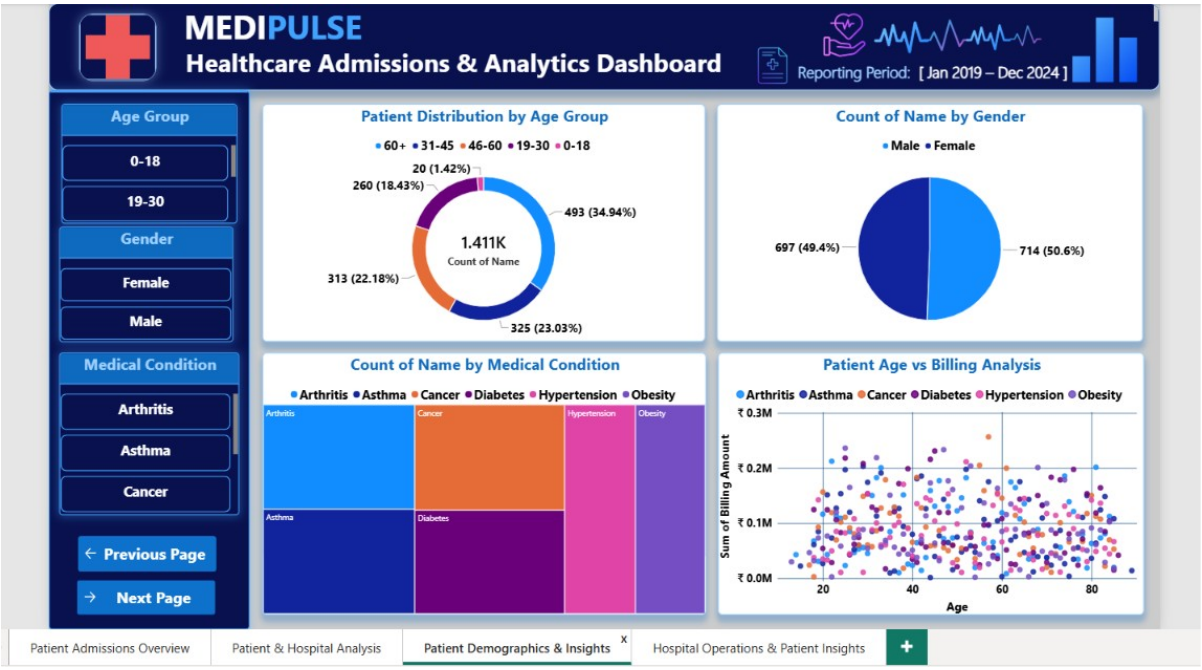


Figure 7: Dashboard 3 – Patient Demographics & Insights.

Dashboard 4 – Hospital Operations & Patient Insights

Total Admissions, Total Hospitals, and Average Age cards; Total Admissions vs Target gauge; Selected 10 Hospitals by Admissions; Hospital-wise Admissions Distribution; Average Patient Age by Hospital. Slicers: Hospital Name, Age, Patient Name.

The Selected 10 Hospitals by Admissions visual uses a manual selection of 10 hospital names. It should not be described as an automatically calculated Top 10 ranking.

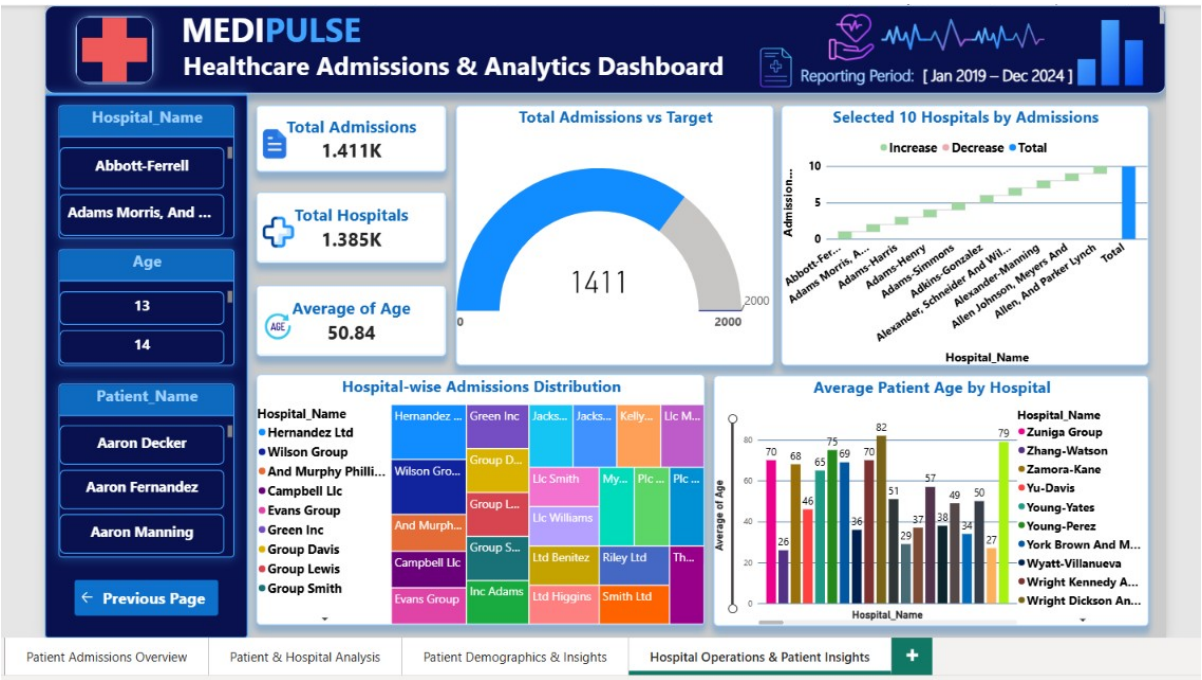


Figure 8: Dashboard 4 – Hospital Operations & Patient Insights.

9. Insights & Conclusions

Visible descriptive results: 1,411 cleaned admission records; 1,385 unique hospital names in the hospital lookup; average patient age approximately 50.84 years; 714 male records (50.6%) and 697 female records (49.4%). The age-group chart shows the 60+ group as the largest group.

Descriptive analytics: summarizes admissions, demographics, hospital counts, and billing totals. Diagnostic analytics: enables comparisons using hospital, age, gender, admission type, and medical condition filters. Predictive analytics: no forecasting model is included. Prescriptive analytics: the dashboards support further investigation but do not generate automated recommendations.

10. Final Conclusion

The MediPulse project demonstrates an end-to-end healthcare data analytics workflow, from preparing admission records to building a relational Power BI model and creating four interactive dashboard pages. Excel and Power Query were used for data preparation, while Power BI and DAX were used to model, analyse, and present the information. The dashboards provide a consolidated view of patient demographics, admissions, hospital activity, and billing patterns.